

Exponentials & Logarithms

ANSWER KEY



Evaluating exponential expressions

- 1 Evaluate 2^5 .
 $2^5 = 32$.
 - 2 Evaluate 3^{-2} .
 $3^{-2} = \frac{1}{9}$.
 - 3 Evaluate $16^{1/2}$.
 $16^{1/2} = 4$.
 - 4 Evaluate $27^{2/3}$.
 $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$.
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Evaluating logarithms

- 5 Evaluate $\log_2 32$.
Since $2^5 = 32$, $\log_2 32 = 5$.
 - 6 Evaluate $\log_3 81$.
Since $3^4 = 81$, $\log_3 81 = 4$.
 - 7 Evaluate $\log_5 1$.
 $\log_5 1 = 0$.
 - 8 Evaluate $\log_{10} 0.01$.
Since $10^{-2} = 0.01$, $\log_{10} 0.01 = -2$.
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Log laws: product and quotient

- 9 Write $\log_2 (8x)$ as a sum of logarithms.
 $\log_2 (8x) = \log_2 8 + \log_2 x = 3 + \log_2 x$.
- 10 Write $\log_3 \left(\frac{x}{9}\right)$ as a difference of logarithms.
 $\log_3 \left(\frac{x}{9}\right) = \log_3 x - \log_3 9 = \log_3 x - 2$.
- 11 Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.
 $\log_5 (5 \times 25) = \log_5 125 = 3$ (since $5^3 = 125$).

- 12 Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.
 $\log_2 \left(\frac{24}{3} \right) = \log_2 8 = 3.$
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Log laws: the power rule

- 13 Write $\log(x^5)$ using the power rule.
 $\log(x^5) = 5\log x.$
- 14 Evaluate $\log_2(4^3)$ using the power rule.
 $\log_2(4^3) = 3\log_2 4 = 3(2) = 6.$
- 15 Write $2\log_3 x + \log_3 y$ as a single logarithm.
 $2\log_3 x + \log_3 y = \log_3(x^2 y).$
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Expanding and condensing logarithmic expressions

- 16 Expand $\log\left(\frac{x^3 y}{z^2}\right)$ fully.
 $3\log x + \log y - 2\log z.$
- 17 Condense $3\ln x - 2\ln y$ into a single logarithm.
 $\ln\left(\frac{x^3}{y^2}\right).$
- 18 Expand $\log_2(\sqrt{x} \cdot y^3)$ fully.
 $\sqrt{x} = x^{1/2}$, so the expression equals $\frac{1}{2}\log_2 x + 3\log_2 y.$
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Solving exponential equations (matching bases)

- 19 Solve $2^x = 32.$
 $32 = 2^5$, so $x = 5.$
- 20 Solve $3^{x+1} = 81.$
 $81 = 3^4$, so $x + 1 = 4 \Rightarrow x = 3.$
- 21 Solve $5^{2x} = 125.$
 $125 = 5^3$, so $2x = 3 \Rightarrow x = \frac{3}{2}.$
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Solving exponential equations using logarithms

- 22 Solve $2^x = 10$ for x , rounded to three decimal places.
 $x = \frac{\log 10}{\log 2} \approx \frac{1}{0.301} \approx 3.322.$

- 23 Solve $5^x = 40$ for x , rounded to three decimal places.
 $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.
- 24 Solve $3^{2x} = 50$ for x , rounded to three decimal places.
 $2x = \frac{\log 50}{\log 3} \approx \frac{1.699}{0.477} \approx 3.561$, so $x \approx 1.781$.
- 25 Solve $e^x = 15$ for x , rounded to three decimal places.
 $x = \ln 15 \approx 2.708$.
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Solving logarithmic equations

- 26 Solve $\log_2(x) = 5$.
 $x = 2^5 = 32$.
- 27 Solve $\log_3(x - 1) = 2$.
 $x - 1 = 3^2 = 9 \Rightarrow x = 10$.
- 28 Solve $\log(x) + \log(x - 3) = 1$.
 $\log[x(x - 3)] = 1 \Rightarrow x(x - 3) = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0$. Reject $x = -2$ (logarithm undefined); $x = 5$.
- 29 Solve $\log_2(x + 3) + \log_2(x - 1) = 5$.
 $\log_2[(x + 3)(x - 1)] = 5 \Rightarrow (x + 3)(x - 1) = 32 \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x + 7)(x - 5) = 0$. Reject $x = -7$ (makes $x - 1 < 0$); $x = 5$.
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Natural logarithms and e

- 30 Evaluate $\ln(e^3)$.
 $\ln(e^3) = 3$.
- 31 Evaluate $\ln(1)$.
 $\ln(1) = 0$.
- 32 Solve $\ln(x) = 2$ for x , in terms of e .
 $x = e^2$.
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Exponential growth and decay models

- 33 The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.
 $P(10) = 5000e^{0.3} = 5000(1.34986) \approx 6749$.
- 34 A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.
 $A(20) = 200e^{-1} = 200(0.36788) \approx 73.6$ grams.
- 35 Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.
Growth — the exponent's coefficient 0.02 is positive, so N increases as t increases.
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Word problems: exponential models in context

- 36 This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.
- a Find the value of the investment after 5 years, rounded to the nearest dollar.
 $A(5) = 2000e^{0.2} = 2000(1.2214) \approx 2443$ dollars.
- b Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$). $2 = e^{0.04t} \Rightarrow 0.04t = \ln 2 \approx 0.693 \Rightarrow t \approx 17.3$ years.
- 37 This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ ($^{\circ}\text{C}$), where t is minutes after brewing and 20°C is room temperature.
- a Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).
 $T(15) = 20 + 60e^{-1.5} = 20 + 60(0.2231) \approx 33.4^{\circ}\text{C}$.
- b Find how many minutes it takes for the temperature to drop to 35°C , rounded to one decimal place (using $\ln 4 \approx 1.386$). $35 = 20 + 60e^{-0.1t} \Rightarrow e^{-0.1t} = 0.25 \Rightarrow -0.1t = -\ln 4 \approx -1.386 \Rightarrow t \approx 13.9$ minutes.
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MCQ — Evaluating exponential expressions

- 38 Evaluate 3^4 .

a. 81 ✓

b. 243

c. 12

d. 64

$3^4 = 81$.

39 Evaluate 4^{-2} .

a. $-\frac{1}{16}$

b. $\frac{1}{16}$ ✓

c. $\frac{1}{8}$

d. 16

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}.$$

40 Evaluate $25^{1/2}$.

a. 5 ✓

b. 50

c. 12.5

d. 625

$$25^{1/2} = \sqrt{25} = 5.$$

41 Evaluate $8^{2/3}$.

a. 5.33

b. 4 ✓

c. 2

d. 16

$$8^{2/3} = (8^{1/3})^2 = 2^2 = 4.$$

42 Evaluate 2^{-3} .

a. -8

b. $\frac{1}{6}$

c. 8

d. $\frac{1}{8}$ ✓

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}.$$

43 Evaluate $32^{3/5}$.

a. 2

b. 16

c. 8 ✓

d. 32

$$32^{3/5} = (32^{1/5})^3 = 2^3 = 8.$$

MCQ — Evaluating logarithms

44 Evaluate $\log_3 27$.

a. 3 ✓

b. 1

c. 27

d. 9

$$3^3 = 27.$$

45 Evaluate $\log_2 64$.

a. 32

b. 8

c. 5

d. 6 ✓

$$2^6 = 64.$$

46 Evaluate $\log_7 1$.

a. 1

b. Undefined

c. 7

d. 0 ✓

$$7^0 = 1.$$

47 Evaluate $\log_{10} 0.001$.

a. 0.001

b. -3 ✓

c. -2

d. 3

$$10^{-3} = 0.001.$$

48 Evaluate $\log_4 16$.

a. 4

b. 8

c. 2 ✓

d. 16

$$4^2 = 16.$$

49 Evaluate $\log_5 125$.

a. 5

b. 3 ✓

c. 1

d. 25

$$5^3 = 125.$$

MCQ — Log laws: product and quotient

50 Write $\log_3 (9x)$ as a sum of logarithms.

a. $2 + \log_3 x$ ✓

b. $2\log_3 x$

c. $9 + \log_3 x$

d. $\log_3 9 + \log_3 x$ only

$$\log_3 (9x) = \log_3 9 + \log_3 x = 2 + \log_3 x.$$

51 Write $\log_5 \left(\frac{x}{25} \right)$ as a difference of logarithms.

a. $\log_5 x - 25$

b. $2 - \log_5 x$

c. $\log_5 x - 2$ ✓

d. $\frac{\log_5 x}{2}$

$$\log_5 \left(\frac{x}{25} \right) = \log_5 x - \log_5 25 = \log_5 x - 2.$$

52 Combine $\log_2 4 + \log_2 8$ into a single logarithm and evaluate.

a. 32

b. 5 ✓

c. 12

d. 2

$$\log_2 (4 \times 8) = \log_2 32 = 5.$$

53 Combine $\log_3 54 - \log_3 2$ into a single logarithm and evaluate.

a. 27

b. 3 ✓

c. 26

d. 52

$$\log_3 (54 \div 2) = \log_3 27 = 3.$$

MCQ — Log laws: the power rule

54 Write $\log(x^7)$ using the power rule.

a. $7\log x$ ✓

b. $x^7\log x$

c. $\log x^7$ unchanged

d. $\log(7x)$

Power rule: $\log(x^n) = n\log x$.

55 Evaluate $\log_3 (9^2)$ using the power rule.

a. 18

b. 81

c. 2

d. 4 ✓

$$2\log_3 9 = 2(2) = 4.$$

56 Write $3\log_2 x + \log_2 y$ as a single logarithm.

a. $3\log_2(xy)$

b. $\log_2(x^3 + y)$

c. $\log_2(x^3y)$ ✓

d. $\log_2(3xy)$

$3\log_2 x = \log_2(x^3)$; combined with $\log_2 y$: $\log_2(x^3y)$.

57 Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$ ✓

b. $\log(4x - y)$

c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

$4\log x = \log(x^4)$; combined: $\log\left(\frac{x^4}{y}\right)$.

MCQ — Expanding and condensing logarithmic expressions

58 Expand $\log\left(\frac{x^2y}{z^3}\right)$ fully.

a. $2\log x + \log y - 3\log z$ ✓

b. $2\log x - \log y - 3\log z$

c. $\log x^2 + \log y - \log z^3$ unchanged

d. $2\log x + \log y + 3\log z$

Apply the quotient and power rules: $2\log x + \log y - 3\log z$.

59 Condense $2\ln x - 3\ln y$ into a single logarithm.

a. $\ln\left(\frac{x^2}{y^3}\right)$ ✓

b. $\ln\left(\frac{x^2}{y}\right) - 3$

c. $\ln\left(\frac{x^3}{y^2}\right)$

d. $\ln(x^2 - y^3)$

$2\ln x = \ln(x^2)$, $3\ln y = \ln(y^3)$: $\ln\left(\frac{x^2}{y^3}\right)$.

60 Expand $\log_2(\sqrt{x} \cdot y^4)$ fully.

a. $\frac{1}{2}\log_2 x + 4\log_2 y$ ✓

b. $\frac{1}{2}\log_2 x + \log_2 y^4$ unchanged

c. $2\log_2 x + 4\log_2 y$

d. $\frac{1}{4}\log_2 x + 2\log_2 y$

$\sqrt{x} = x^{1/2}$: $\frac{1}{2}\log_2 x + 4\log_2 y$.

61 Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.

a. $\log(x \cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x} \cdot y^2)$ ✓

d. $\log(2\sqrt{x} \cdot y)$

$\frac{1}{2}\log x = \log \sqrt{x}$; combined: $\log(\sqrt{x} \cdot y^2)$.

MCQ — Solving exponential equations (matching bases)

62 Solve $2^x = 64$.

a. $x = 8$

b. $x = 6$ ✓

c. $x = 32$

d. $x = 5$

$2^6 = 64$.

63 Solve $3^{x+2} = 27$.

a. $x = 9$

b. $x = 1$ ✓

c. $x = 5$

d. $x = 3$

$3^{x+2} = 3^3 \Rightarrow x + 2 = 3 \Rightarrow x = 1$.

64 Solve $5^{3x} = 125$.

a. $x = \frac{1}{3}$

b. $x = 41.67$

c. $x = 3$

d. $x = 1$ ✓

$5^{3x} = 5^3 \Rightarrow 3x = 3 \Rightarrow x = 1$.

65 Solve $4^x = \frac{1}{16}$.

a. $x = 4$

b. $x = -4$

c. $x = -2$ ✓

d. $x = 2$

$4^x = 4^{-2} \Rightarrow x = -2$.

66 Solve $2^{2x-1} = 32$.

a. $x = 3$ ✓

b. $x = 2.5$

c. $x = 6$

d. $x = 5.5$

$$2^{2x-1} = 2^5 \Rightarrow 2x - 1 = 5 \Rightarrow x = 3.$$

MCQ — Solving exponential equations using logarithms

67 Solve $2^x = 12$ for x , rounded to three decimal places.

a. 3.585 ✓

b. 3.170

c. 4.000

d. 2.485

$$x = \frac{\ln 12}{\ln 2} \approx 3.585.$$

68 Solve $7^x = 50$ for x , rounded to three decimal places.

a. 7.143

b. 2.010 ✓

c. 3.000

d. 1.505

$$x = \frac{\ln 50}{\ln 7} \approx 2.010.$$

69 Solve $4^{2x} = 90$ for x , rounded to three decimal places.

a. 6.492

b. 3.246

c. 1.623 ✓

d. 0.811

$$2x = \frac{\ln 90}{\ln 4} \approx 3.246 \Rightarrow x \approx 1.623.$$

70 Solve $e^x = 25$ for x , rounded to three decimal places.

a. 25.000

b. 2.303

c. 3.219 ✓

d. 12.500

$$x = \ln 25 \approx 3.219.$$

MCQ — Solving logarithmic equations

71 Solve $\log_3(x) = 4$.

a. $x = 7$

b. $x = 12$

c. $x = 64$

d. $x = 81$ ✓

$$3^4 = 81.$$

72 Solve $\log_2(x - 1) = 3$.

a. $x = 6$

b. $x = 9$ ✓

c. $x = 7$

d. $x = 8$

$$x - 1 = 2^3 = 8 \Rightarrow x = 9.$$

73 Solve $\log(x) + \log(x - 3) = 1$.

a. $x = -2$

b. $x = 5$ ✓

c. $x = 10$

d. $x = 13$

$$\log(x(x - 3)) = 1 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0; \text{ only } x = 5 \text{ keeps both logs defined.}$$

74 Solve $\log_2(x + 2) + \log_2(x - 2) = 5$.

a. $x = 6$ ✓

b. $x = -6$

c. $x = 34$

d. $x = 18$

$$(x + 2)(x - 2) = 2^5 = 32 \Rightarrow x^2 - 4 = 32 \Rightarrow x^2 = 36; \text{ only } x = 6 \text{ keeps both logs defined.}$$

MCQ — Natural logarithms and e

75 Evaluate $\ln(e^5)$.

a. 0

b. e^5

c. 1

d. 5 ✓

$$\ln(e^x) = x.$$

76 Evaluate $\ln\left(\frac{1}{e}\right)$.

a. 1

b. e

c. -1 ✓

d. 0

$$\ln(e^{-1}) = -1.$$

77 Solve $\ln(x) = 3$ for x , in terms of e .

a. $x = e^{1/3}$

b. $x = 3$

c. $x = e^3$ ✓

d. $x = 3e$

$$\ln(x) = 3 \Rightarrow x = e^3.$$

78 Evaluate $e^{\ln 7}$.

a. 7 ✓

b. $\ln 7$

c. 1

d. e^7

$$e^{\ln x} = x.$$

MCQ — Exponential growth and decay models

79 The population of a town is modeled by $P(t) = 8,000e^{0.025t}$, where t is in years. Find $P(15)$, rounded to the nearest whole number.

- a. 8,300
- b. 11,000
- c. 11,640 ✓
- d. 12,000

$$P(15) = 8,000e^{0.375} \approx 11,640.$$

80 A radioactive substance decays according to $A(t) = 150e^{-0.04t}$ (grams). Find $A(25)$, rounded to one decimal place.

- a. 60.7
- b. 50.0
- c. 75.0
- d. 55.2 ✓

$$A(25) = 150e^{-1} \approx 55.2 \text{ g.}$$

81 Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

- a. Decay ✓
- b. Neither
- c. Growth
- d. Cannot be determined

The exponent's coefficient is negative: decay.

82 Determine whether $M(t) = 30e^{0.01t}$ models growth or decay.

- a. Neither
- b. Cannot be determined
- c. Decay
- d. **Growth ✓**

The exponent's coefficient is positive: growth.

83 The graph shows an exponential function. Based on the graph, does it represent growth or decay?

- a. Neither
- b. **Growth ✓**
- c. Cannot be determined
- d. Decay

The curve rises as x increases: growth.

MCQ — Word problems: exponential models in context

84 An investment of 3,000 dollars grows according to $A(t) = 3,000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

- a. 4,200
- b. 4,800
- c. **4,475 ✓**
- d. 3,400

$A(8) = 3,000e^{0.4} \approx 4,475$ dollars.

- 85 A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ ($^{\circ}\text{C}$), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

- a. 38.5°C
- b. 50.0°C
- c. 43.1°C ✓
- d. 22.0°C

$$T(15) = 22 + 70e^{-1.2} \approx 43.1^{\circ}\text{C}.$$

- 86 A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

- a. 4,000 ✓
- b. 2,000
- c. 8,000
- d. 1,500

$$P(9) = 500 \times 2^3 = 4,000.$$

- 87 The value of a car depreciates according to $V(t) = 25,000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

- a. 11,093 ✓
- b. 12,750
- c. 9,500
- d. 10,625

$$V(5) = 25,000(0.85)^5 \approx 11,093 \text{ dollars}.$$